

ERTH 467 – Time Series Analysis and 1D Signal Processing (F2026) Lab Assignment #2

In this lab assignment, you will develop a code to filter noisy data with band-pass, prediction and matched filters. The data for this assignment is provided in the files:

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“erth_467_lab_2_data.mat”,  
“erth_467_lab_2_data.txt”,  
“erth_467_lab_2_wavelet.txt”
```

on D2L (the files contain the same data, use the **.mat** file if you are coding in MATLAB and the **.txt** files otherwise). Based on your results, you will write a brief scientific report (see report format and rubric under Lab Information on D2L) discussing your results. You will submit the assignment to the Dropbox on D2L as a compressed (**.zip**) file entitled **“ERTH_467_Lab_2_Name.zip”** containing a copy of your report in **.pdf** format and a copy of all code used to generate your report results. In the file name, replace Name with your surname.

The data file contains measured data in **“data”**, the associated times in **“t”**, the source wavelet in **“wavelet”**, and the associated times in **“tw”**. In the text files, the first and second columns of **“...data.txt”** contain the measured data and measurement times, while the first and second columns of **“...wavelet.txt”** contain the wavelet and associated times. The data were generated using a random reflectivity series, white noise, and the provided wavelet.

Key MATLAB functions for this lab are **fft**, **ifft**, **fftshift**, and **ifftshift**. The function **fft** (and **ifft**) applies the (inverse) fast Fourier transform. By default, the transformed result from **fft** is in the frequency ordering $[0 : df : (f_{nyq} - df), -f_{nyq} : df : -df]$. This is usually not desirable for analysis. The function **fftshift** re-arranges the transform result to the frequency ordering $[-f_{nyq} : df : f_{nyq} - df]$, which is more naturally arranged from low to high frequencies. To work with this ordering, we will usually call the forward and inverse Fourier transforms with $a = \text{fftshift}(\text{fft}(b))$ and $b = \text{ifft}(\text{ifftshift}(a))$. Familiarize yourself with these functions before moving on. It may be worthwhile to create functions **t_from_f** and **f_from_t** to determine the times associated with a set of frequencies and vice versa.

1. Write a function to perform the convolution of two vectors via frequency domain multiplication. Make sure to allow for the possibility of vectors of different lengths. Verify that your function works by using it to convolve the provided wavelet with a delta function time series. Make sure that the maximum value of the convolution result occurs at the same time as the delta spike.
2. Based on the information in the wavelet, design both Ormsby and Butterworth band-pass filters to improve the signal-to-noise ratio. Find the parameters for both types of filter that seem best for this data set. Create a figure with subplots showing the original data, the Ormsby-filtered result and the Butterworth filtered result. Choose your preferred filtered result; this will be your “band-passed data”. Explain why you chose the parameters you did.
3. Using the first half of the measured data, design one prediction filter of length 30 and one prediction filter of length 2000. Apply these filters to the full measured data (including all times now). Which filter does a better job of predicting the first half of the measured data? Which does a better job of predicting the first half of the band-passed data? Which of these data sets did we design the filter to optimally match? Which do we want the filter to match? How does the quality of the prediction change for the second half of the data (which we did not use in designing the filter)? What does this say about generalizability? How does the better of the prediction filtering results compare to the band-passed data? What advantages does this filter have in terms of prior knowledge? Create a figure with subplots showing the original data, the band-passed data, the

short filter result and the long filter result.

4. Design a matched filter to maximize the response to the input wavelet, then apply it to the measured data. How does the result compare to the band-passed data? How much prior information was needed to design this filter? Create a figure with subplots that show the original data, the band-passed data, and the matched-filtered data.
5. Repeat steps 3 and 4, designing and applying the filters with the band-passed data rather than the original measured data. How do the results change? What does this say about the robustness of the filters used to white noise?

Write a brief report on your results following the format posted on D2L. The report and code (together in a compressed file) should be submitted to the Dropbox on D2L by 23:30 on Friday, October 17, 2025.